

COMPUTER ENGINEERING CAPSTONE PROJECT · ÇANKAYA UNIVERSITY

VR Anatomy

VR-Based Educational Interactive Human Anatomy Training Platform

Team Members:

Elifnaz Talas
Kutay Ayoğlu

Öykü Çoban
Çağla Pelin Doğan

Sena Akbaba

Supervisor:

Assoc. Prof. Dr. Gül Tokdemir
Dr. Instructor Talha Karadeniz



Supported by TÜBİTAK 2209-A



MAIN PROBLEM

Anatomy Education Lacks Depth

2D Diagrams Limit Spatial Understanding

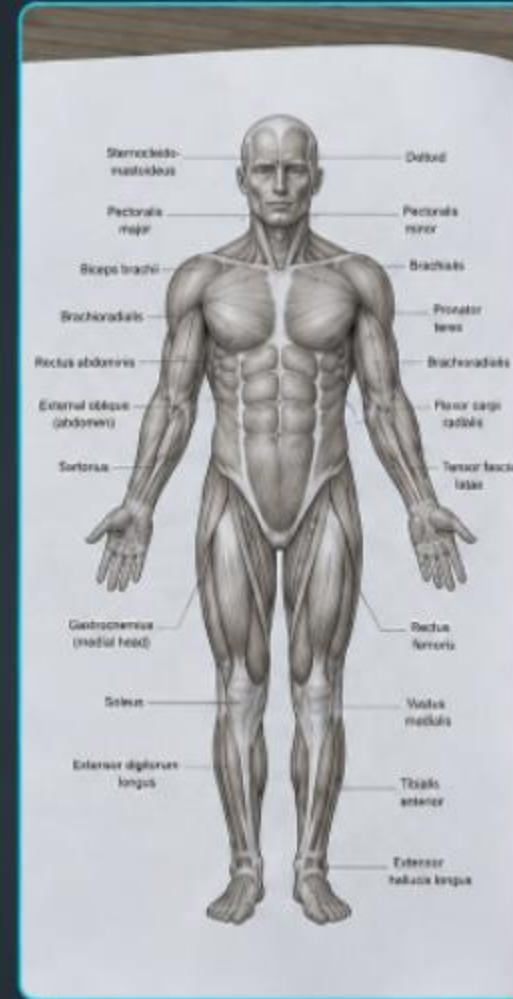
Complex 3D body structures taught through flat textbook illustrations

Physical Labs Are Costly and Not Scalable

Cadaver labs and physical models unavailable at most high schools

Students Read Anatomy — They Don't Explore It

Passive consumption, no interactive engagement with structures



Traditional



VR Anatomy

Target Audience & Scope



Target Audience
9th-Grade Students



PILOT SCHOOL

**Lokman Hekim Vocational and
Technical Anatolian High School**



Movement System

Bones & Joints

Muscles

3D Spatial Exploration



Circulatory System

Heart Anatomy

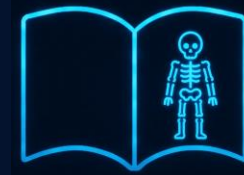
Major Vessels

Six Core Goals



Immersive 3D Exploration

VR-native anatomical model interaction



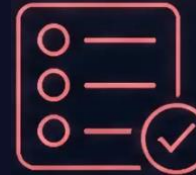
MEB Curriculum Alignment

Ministry of Education anatomy syllabus



On-Demand AI Tutor

Anatomy Q&A available during every session



Quiz-Based Assessment

Immediate feedback after learning units



User-Friendly VR Interface

Designed for first-time VR users



Modular Architecture

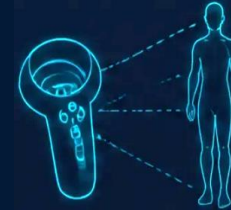
Unity client + FastAPI backend separation

Four Learning Modes

AI Guided Learning



Free Explore



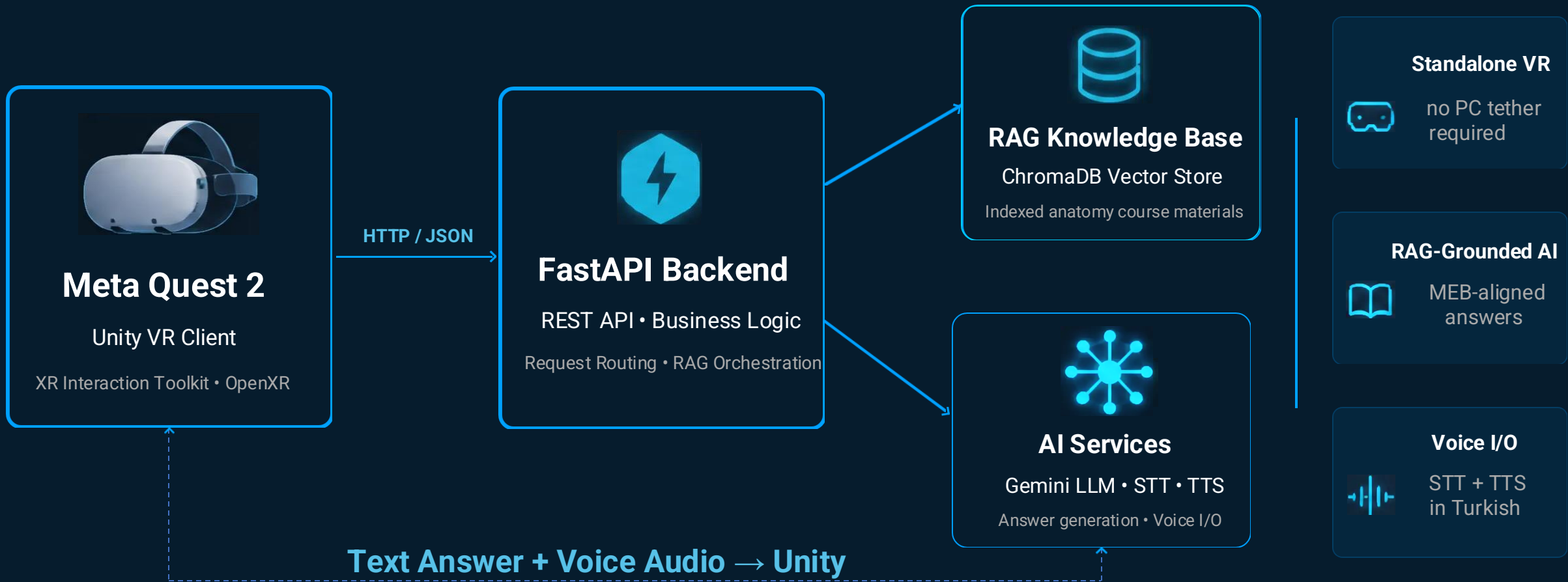
Quiz



AI Tutor



Client-Server Architecture



Engineering Trade-offs



Cloud vs. Local RAG

We used a local ChromaDB index to reduce costs and have more control over retrieval.



Gemini Over GPT-4o

Gemini offered a generous free tier and strong Turkish output — letting the team iterate without subscription overhead during development.



Hand-Tracking vs. Controllers

XR Hands enables bare-hand interaction, but controllers stay active — many users prefer them for tight bone clusters.



Quiz Performance Feedback

Green / Yellow / Red feedback makes quiz results Easy to understand for 9th-grade students.

Pilot Testing — Pre/Post Methodology

42 Students

Participants

Lokman Hekim MTAL

Pilot School

Pre-test → Post-test

Methodology

Formal Style

22 students · May 13, 2026



Friendly Style:
43% higher
improvement

Friendly Style

20 students · May 20, 2026



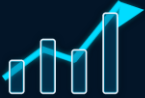
Pre-test Average

Post-test Average

Key Outcomes

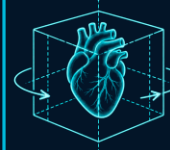
Project Impact

IMPACT



Learning improvement observed

Higher post-test scores after the VR session.



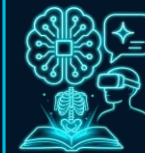
Better spatial understanding

3D inspection supported bones and heart learning.



Pilot testing validated usability

Students used the system in a real classroom evaluation.



AI support added value

Instant, simpler explanations helped guide review.



Higher engagement

VR changed anatomy from reading into active exploration.



Expandable learning platform

Modular design can grow into more anatomy systems.

Outcome focus: measurable learning feedback + active 3D anatomy + AI-guided support

Future Works

Next Iteration Roadmap



1 · Improve Learning Loop

- Connect weak quiz topics to guided review scenes
- Add teacher dashboard and progress analytics
- Refine AI Tutor accuracy, latency, STT/TTS quality



2 · Expand Content & Interaction

- Add more anatomy systems beyond current scope
- Include animated physiology flows and model labels
- Improve hand tracking, haptics, and object manipulation



3 · Validate at Larger Scale

- Run larger classroom studies with more schools
- Use delayed post-tests to measure retention
- Compare formal vs. friendly guidance with more data

Long-term vision: a scalable, curriculum-aligned VR anatomy lab for schools with limited physical lab access

Demo Time





THANK YOU

ANY QUESTIONS?

